

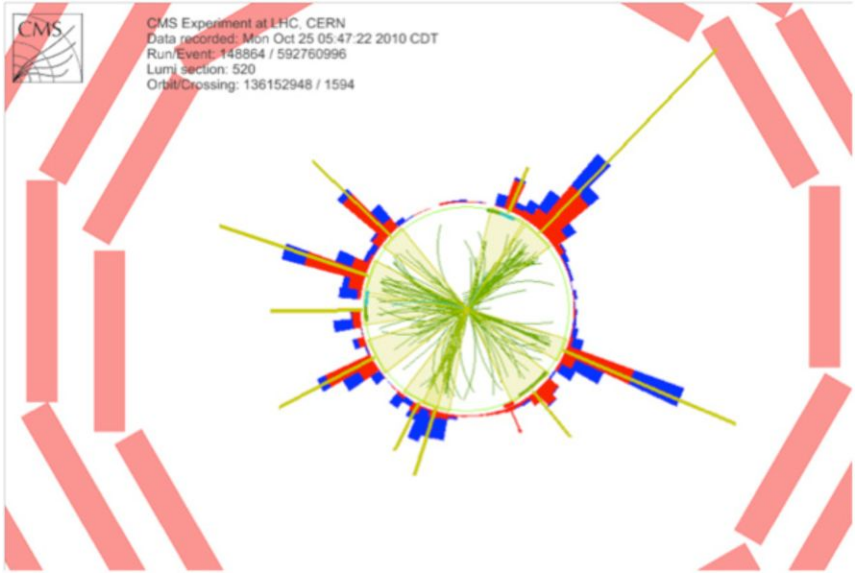
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POLSKIEJ AKADEMII NAUK

Beauty in the shower: decoding b-quark jet substructure

José Fernando Gomez Bedoya , Anton Bukhtatyi

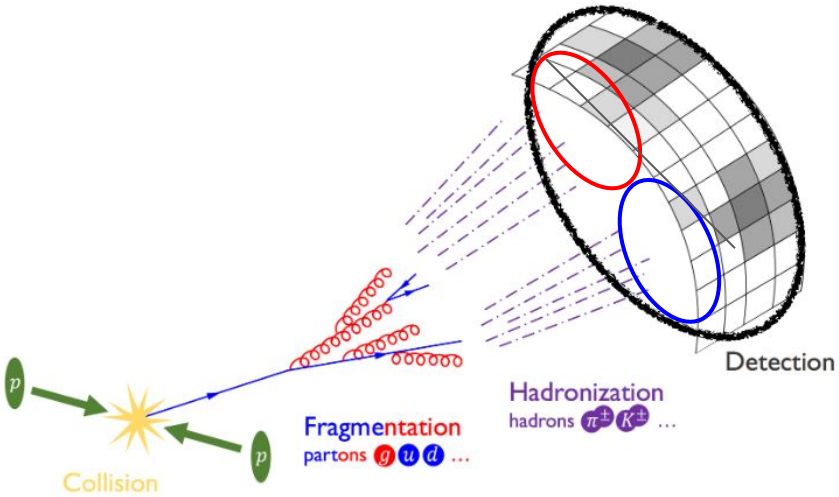
Tutor: Agnieszka Ogrodnik

This project investigates the properties of jets reconstructed in Monte Carlo simulations and proton–proton collision data recorded by the ATLAS experiment. The study aims to optimise the Graph Neural Network (GN) discriminant for efficient b-jet identification and to compare the characteristics of simulated and experimental jets. Particular emphasis is placed on the study of jet substructure through the reconstruction of sub-jets and the analysis of the angular separation between them, providing further insight into the partonic origin and internal structure of reconstructed jets.



https://cms.cern/news/search-microscopic-black-hole-signatures-large-hadron-collider

What are jets?



https://indico.phys.ethz.ch/event/37/contributions/320/attachments/252/396/analysis_techniques_HEP.pdf

A jet is a collimated spray of many particles, mostly hadrons.

They are created in high energy collisions, when energetic partons (quarks and gluons) are produced.

These undergo fragmentation, where they radiate additional partons through perturbative QCD.

As these partons lose energy, they hadronize, combining into color-neutral hadrons (π^+, π^-, π^0, K^+, …)

By studying jets, we recover information about the high-energy partons created in collisions

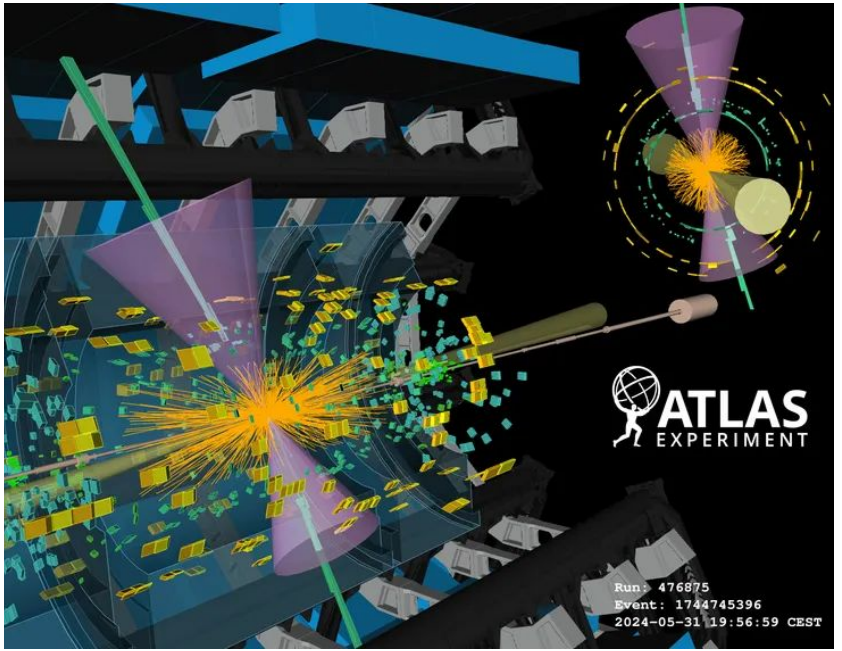
Why are we interested in jets?

Key areas where jets play a central role:

- *Higgs and electroweak physics, especially in vector boson fusion and scattering*
- *Searches for new physics (BSM), including dark matter signatures*
- *Studying QCD, such as probing parton distribution functions (PDFs) and measuring the strong coupling constant (α□)*

Goal of our study:

- *Heavy ions collision, text*



https://atlas.cern/Updates/Briefing

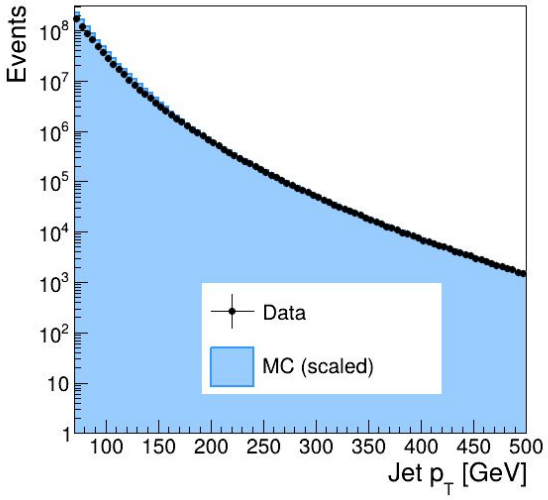
MC vs Experiment

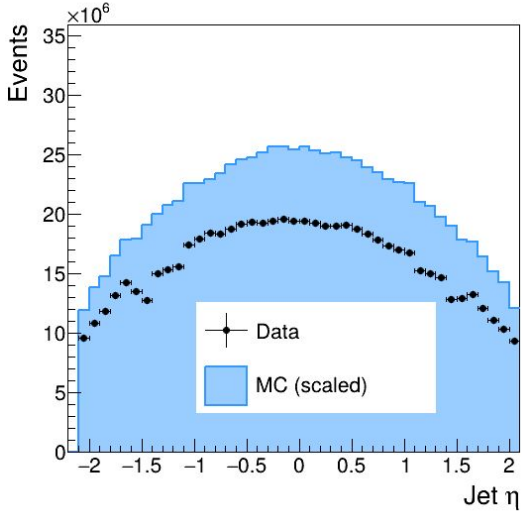
Experimental data:

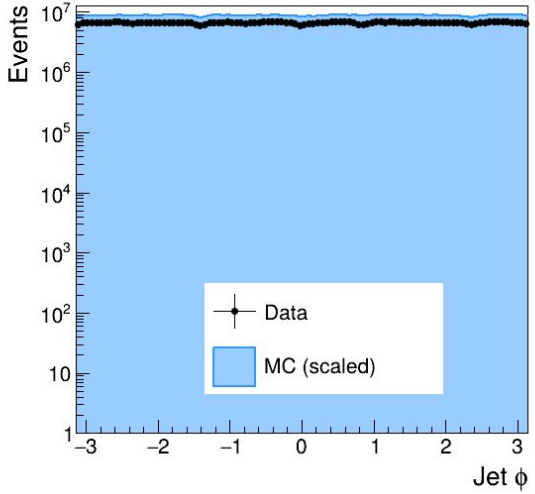
Collected from *pp* collisions in 2024 at center of mass energy of 5.36 TeV, with a *pT* > 70 GeV cut

Simulated event data:

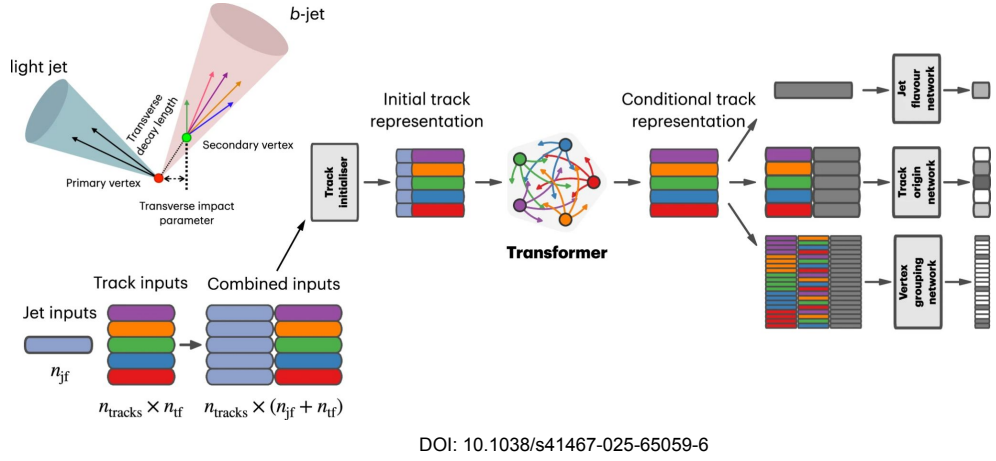
Generated in PYTHIA 8 at the same center of mass energy, same *pT* cut, and |η|<2.1 cut



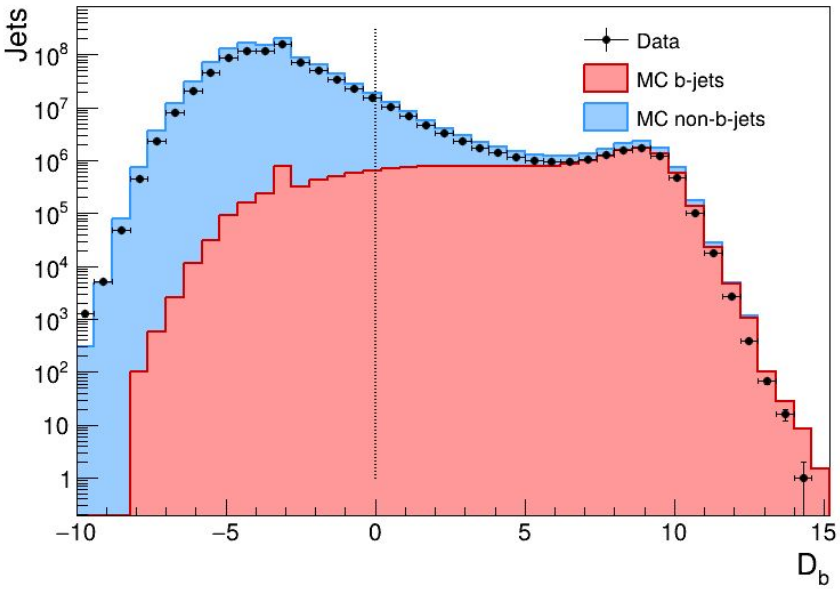




Graph Neural Network (GN) discriminant



DOI: 10.1038/s41467-025-65059-6



GN discriminant:

$$D_b = \log\left(\frac{p_b}{f_c \cdot p_c + f_\tau \cdot p_\tau + (1 - f_c - f_\tau) \cdot p_u}\right)$$

where: $f_\tau = 0.01 \quad f_c = 0.2$

p_i *-probability that this jet is i-jet*

Graph Neural Network (GN) discriminant

Efficiency correction factor:

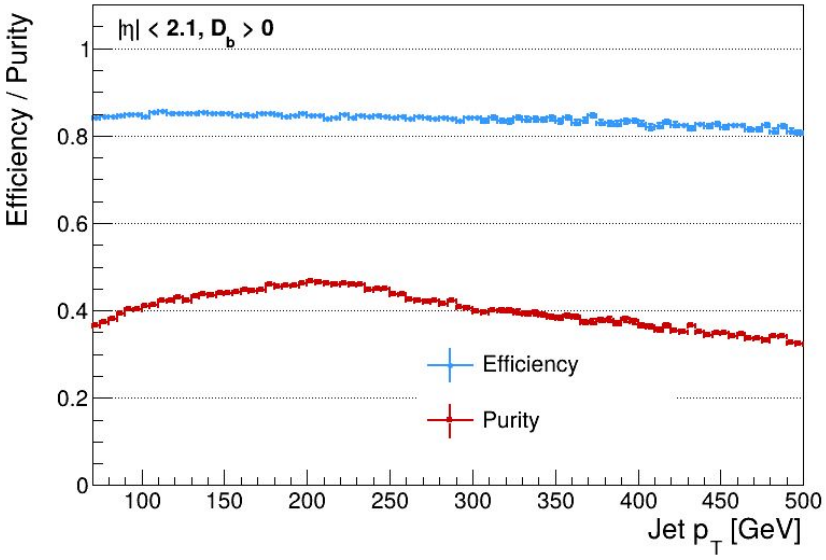
$$\varepsilon = \frac{\text{number of b-jets with } D_b > D_b \text{ cut}}{\text{number of all b-jets}}$$

Contamination correction:

$$p = \frac{\text{number of b-jets with } D_b > D_b \text{cut}}{\text{number of all jets with } D_b > D_b \text{ cut}}$$

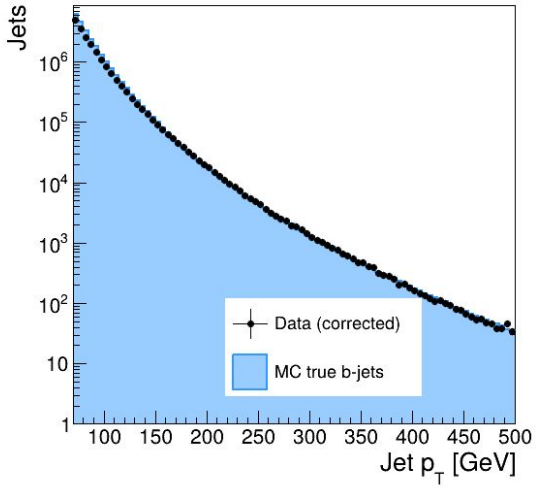
Scaling factor:

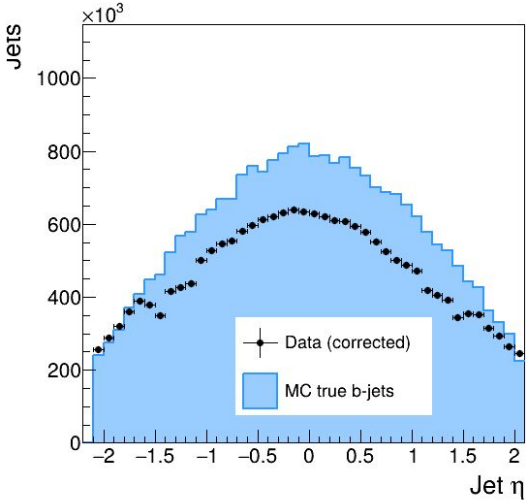
$$sc = \frac{p}{\varepsilon}$$

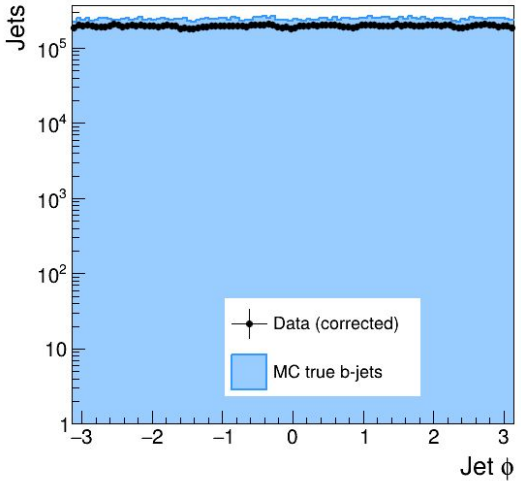


Corrected D_b selected distribution

Reco jets in data passing the GN2 discriminant cut (D_b> 0) are corrected bin-by-bin using *pT*-dependent efficiency (ε) and purity (p) derived from MC:

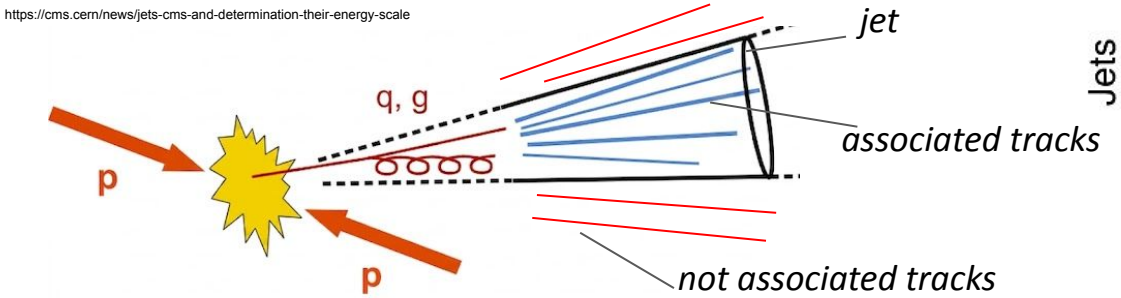


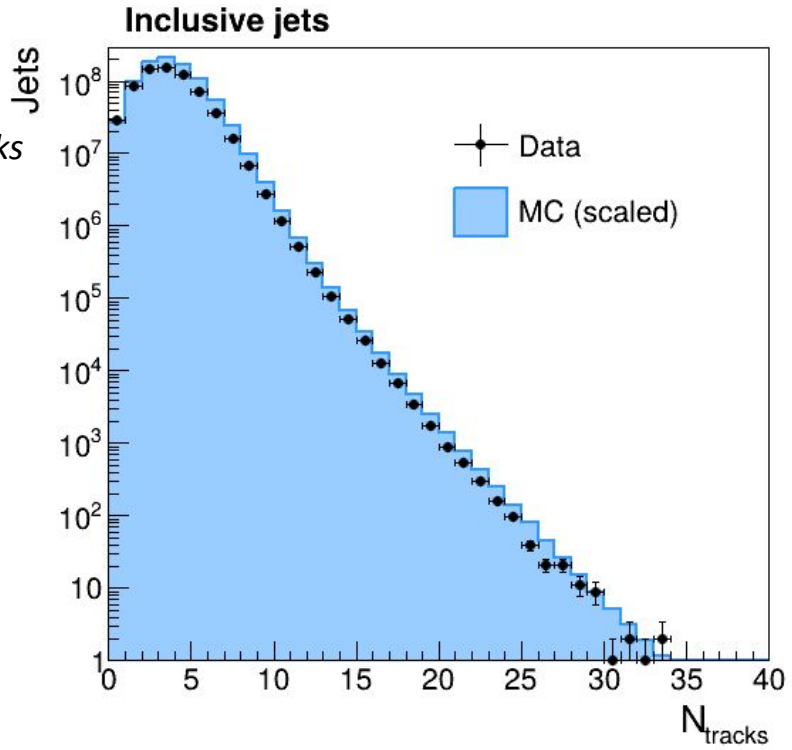




Ghost jet association

Which tracks are associated with this jet?





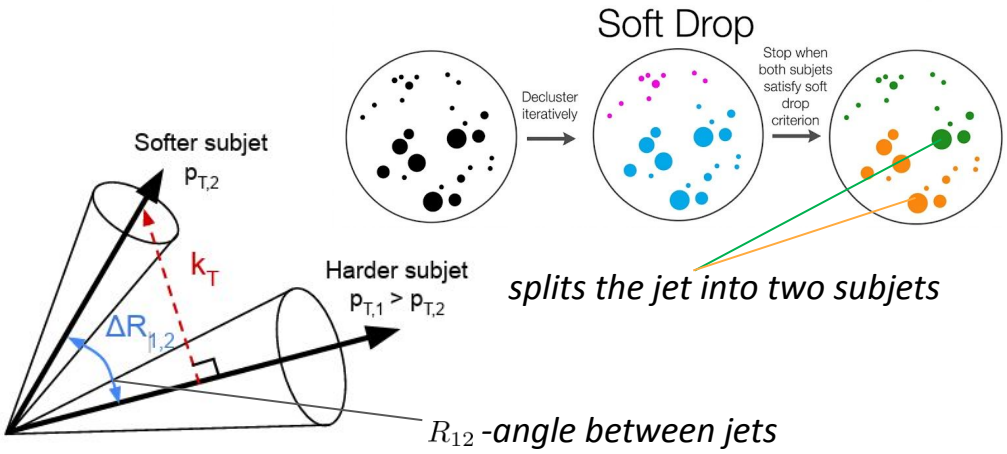
Ghost association method:

1. *Create ghost tracks by assigning each track a very small (p_T) and representing it as a single ghost vector.*
2. *Add the jet as a component to the current set of ghosts.*
3. *Run the anti-(k_t) clustering algorithm.*
4. *Recover the (p_T) of the tracks associated with the identified ghosts.*

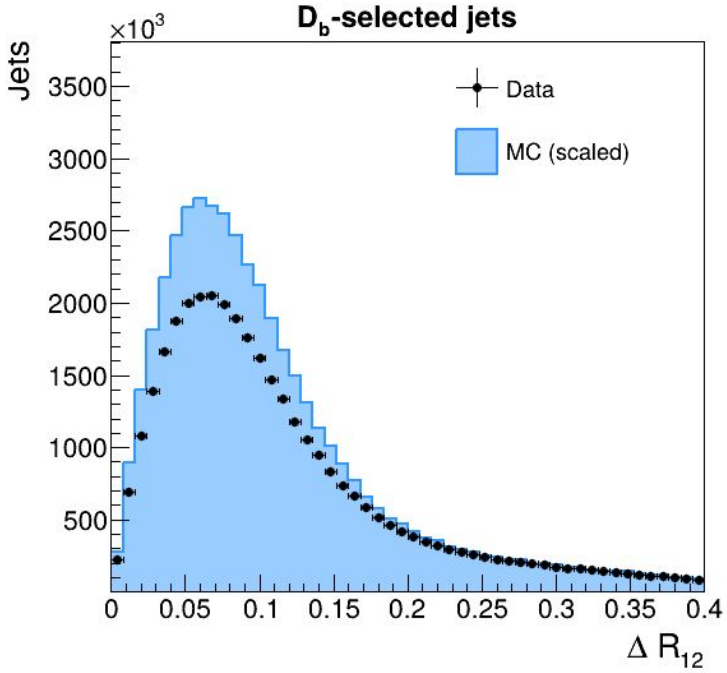
Substructure studies

Scheme of the angle determination between subjets:

Ghost jet association → Selected tracks → Cambridge algorithm + Soft Drop → Angle between subjets

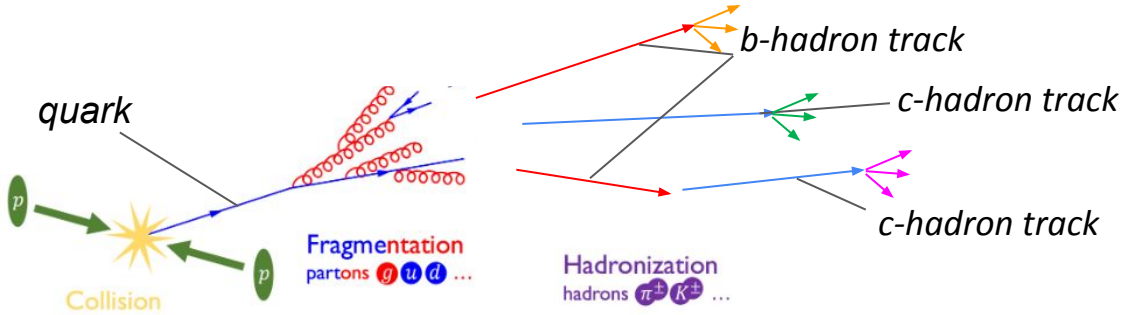


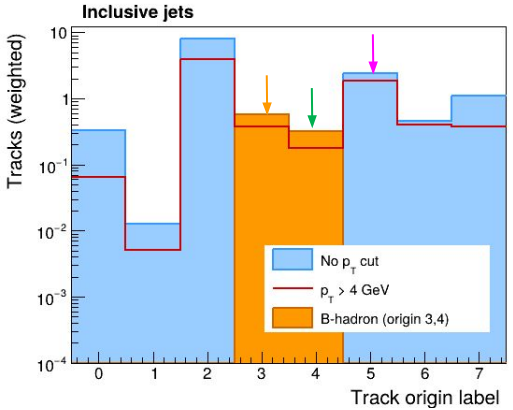
splits the jet into two subjets

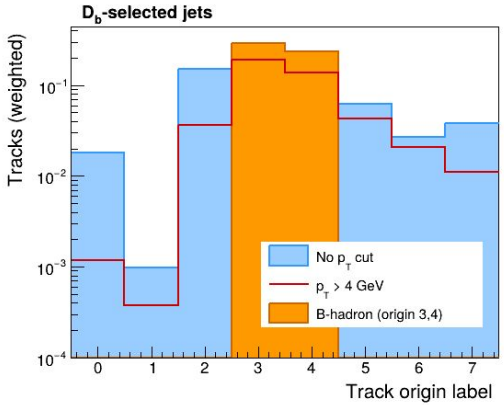


Track origin

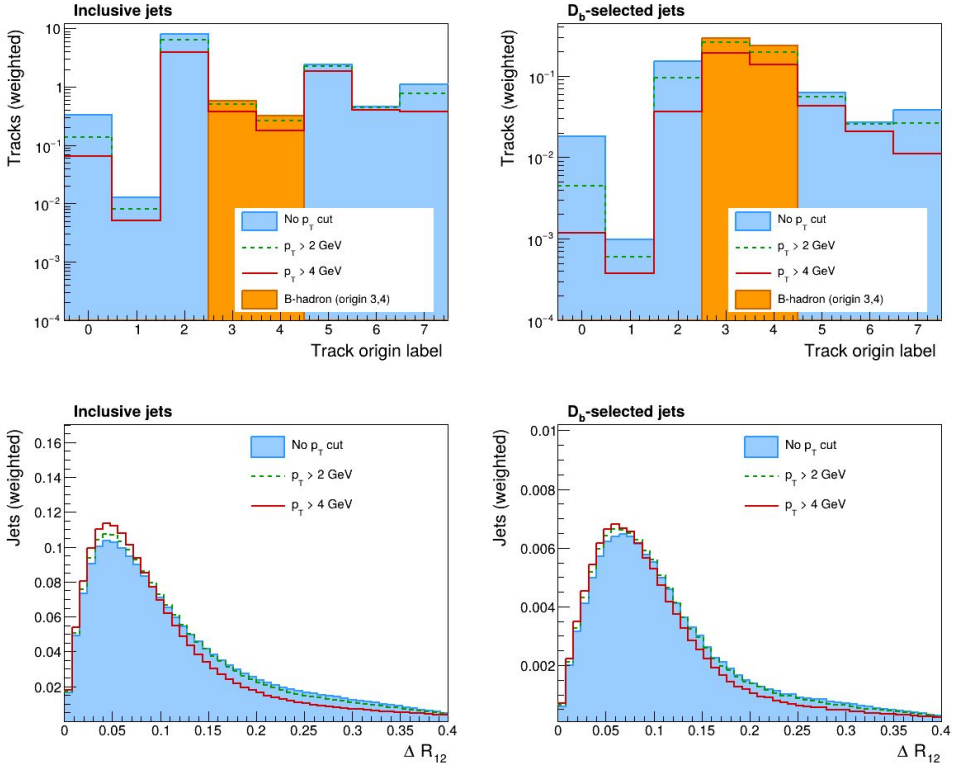
Proton proton collision:







Transverse momentum cuts



Track merging

